

ADAPTING TO ENVIRONMENTAL RISK: EVIDENCE FROM THE U.S. WEST COAST'S MOST VALUABLE FISHERY

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MOTIVATION

A central question in environmental and resource economics is how quickly firms adjust to changing ecological and environmental conditions.

Climate change makes this question especially important for fisheries:

- marine species are shifting across space;
- climate anomalies are increasing ecological disruptions;
- harmful algal blooms create sudden harvest and market risks.

In fisheries, adaptation is shaped not only by harvesters, but also by the institutions that govern access to the resource.

This paper studies one adaptive policy instrument: evisceration orders in Oregon's Dungeness crab fishery.

CLIMATE CHANGE AND FISHERIES RISK

Climate change is reshaping marine ecosystems and challenging existing fisheries governance (Pinsky et al., 2018).

Along the U.S. West Coast, marine heatwaves and climate anomalies have been linked to severe blooms of *Pseudo-nitzschia*, the diatom that produces domoic acid (Trainer et al., 2020).

Domoic acid accumulates in crab viscera and can force delays, closures, or modified harvest rules.

Emerging evidence suggests climate variability affects the frequency, intensity, and spatial extent of domoic-acid-producing harmful algal blooms (Kessouri et al., 2026).

Climate anomaly → HAB → Domoic acid event → Management response



Sea lion suffering involuntary muscle spasm from domoic acid
(National Oceanic and Atmospheric Administration, 2025)

FROM CLOSURES TO ADAPTIVE MANAGEMENT

Fisheries managers often face a binary choice:

Open fishery vs. **Full closure**

Climate-ready fisheries management emphasizes policies that preserve economic activity while maintaining conservation and public-health objectives (?).

Evisceration orders are an intermediate policy:

- harvest can continue;
- contaminated viscera must be removed;
- access to the premium live/whole crab market is eliminated.

Key question: How do firms(harvesters) react to this adaptive policy?

EVISCERATION ORDERS IN OREGON

Trigger: domoic acid in Dungeness crab viscera exceeds the regulatory threshold.

Policy response: crab harvest may continue only if viscera are removed before sale.

Economic mechanism:

- avoids a full closure;
- removes access to live and whole-crab markets, live crab is more valuable than eviscerated/processed crab (Smith, 2023)

Why this matters: the policy preserves activity, but may redistribute costs toward vessels, ports, or buyers with less flexibility.

THE COUNTERFACTUAL

Closure vs. Evisceration

The alternative to an evisceration order is not a normally operating fishery. It is often a full closure with little or no harvest revenue.

Prior work estimates that full closure would reduce expected revenue by 98.5 percent (Wu et al., 2026).

So evisceration should be evaluated as a partial adaptation policy:

It prevents closure, but changes the terms of market

The empirical question is whether the policy mainly disrupts fishing behavior or compresses revenue through lower prices.

WHY THIS SETTING?

Dungeness crab is the most valuable fishery on the U.S. West Coast.

Oregon is an especially useful setting because:

- it was the first state to authorize evisceration orders;
- orders vary across zones and timing;
- the fishery is a limited-entry derby;
- most catch occurs in the first 6–8 weeks.

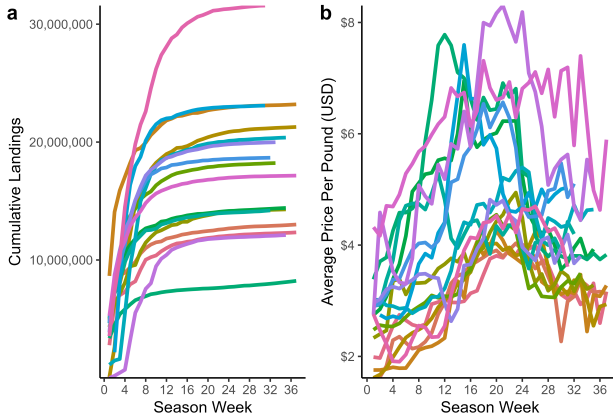
This compressed season sharply limits short-run adjustment margins.

Interpretation: if vessels do not exit or relocate, the policy burden is likely transmitted through prices.

THE DERBY STRUCTURE LIMITS ADJUSTMENT

The Dungeness crab fishery is a limited-entry derby: vessels race to harvest before the season's value is exhausted (Deweese et al., 2004).

Cumulative Landings and Average Prices by Season Week



Panel A: Cumulative Landings Most catch occurs early, leaving little time to adapt.

Panel B: Average Price
Evisceration removes access to the premium market channel.

Prediction

Short-run supply should be inelastic; prices should adjust more than effort.

IDENTIFICATION STRATEGY

Variation exploited. Spatial and temporal variation in evisceration order implementation across Oregon management zones, 2016–2023.

Design. Difference-in-differences with a stacked event study extension (Cengiz et al., 2019; Wing et al., 2024).

Outcomes. Ex-vessel prices, landings, revenue, effort, vessel participation, and spatial reallocation of fishing activity.

ORDERS

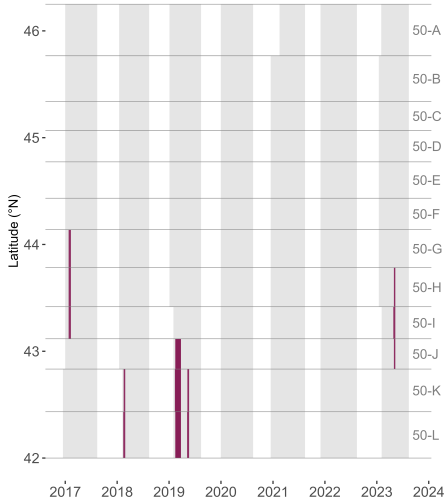
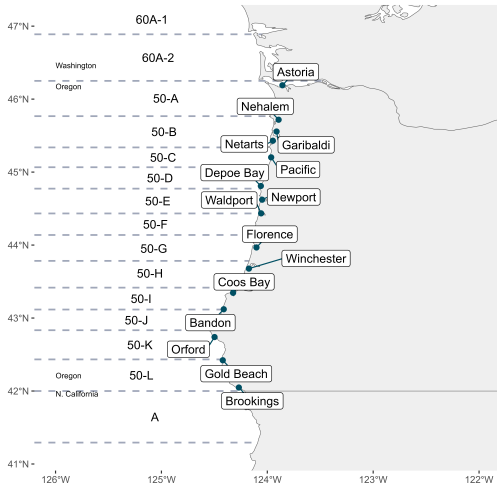


Figure on left depicts management areas and ports. Right figure depicts seasons and policy periods

ORDERS

Orders are: spatially targeted (zone-specific, not coastwide); short-lived (average 12–18 days); staggered across seasons and zones.

Treatment is the area-week cell under an evisceration order, either as a violative or buffer zone. Control is all area-week cells in the same season not subject to an order.

DATA

Unique access to detailed private-firm level microdata

- Vessel-level fishing trip decisions, precise catch and revenue records.
- High-frequency monitoring: daily/weekly spatial decisions captured through VMS tracking and detailed logbook

Rich empirical setting allows examination of individual firm behavior under environmental policy.

DATA

Logbooks (ODFW, 2022)

Self-reported string-level records: start/end coordinates, pots set, soak duration, estimated catch. 625 unique vessels, 2007–2023. Limited-spatial information.

Fish Tickets (ODFW, 2022)

Mandatory dealer records: weight, price, date, port, vessel characteristics. Matched to logbooks via ticket number.

VMS (NMFS OLE)

Hourly vessel position for 406 vessels. 25M raw observations filtered to 103,780 trips. Used to verify logbook string locations and measure fishing hours.

Domoic Acid Testing (ODA)

Bi-monthly testing by zone. Separate measures for crab meat and viscera. Bivalve contamination data included as an early-warning indicator.

DATA GENERATION



Logbooks provide spatial information: GPS coordinates of vessel activity, dates, port of landing

Logbooks report “strings”—start and end points of crab pots

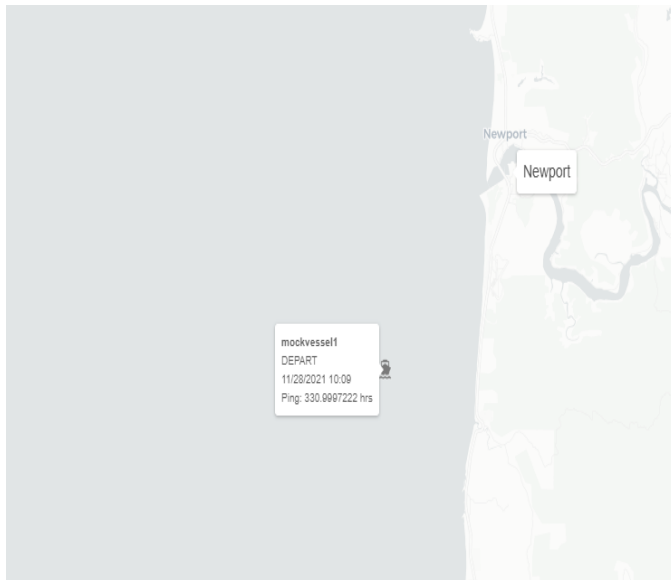


Fish tickets provide information on amount of crab caught and revenues



Data Set is constructed by using logbook data to simulate pots along each string and intersecting them to each harvest area—summing up total number of pots per area proportional to each string’s reported total

DATA EXAMPLE



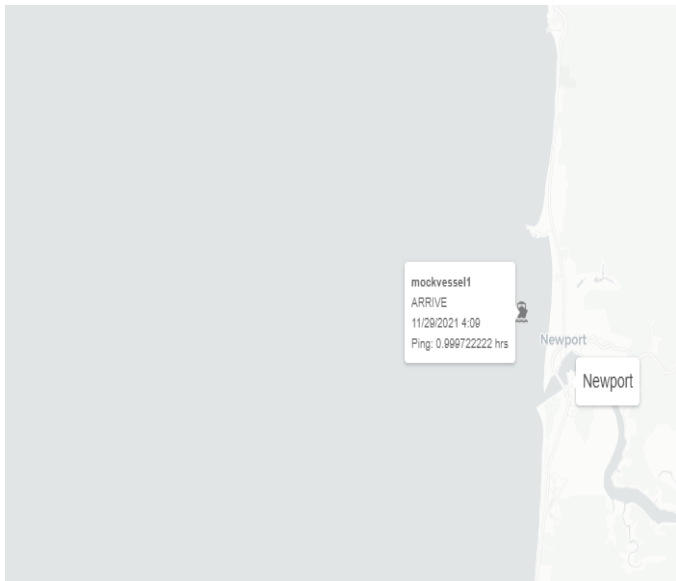
DATA EXAMPLE



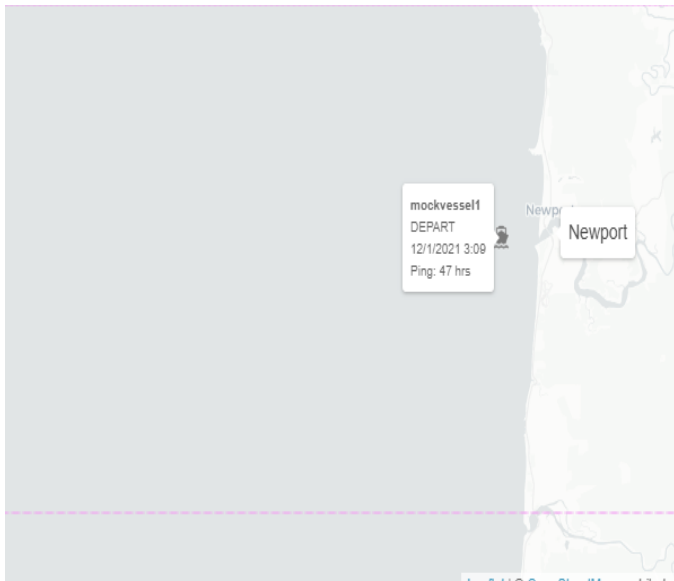
Log Detail ID: 253343 | Date: 11/28/2021 | Vess: mockvessel1

mockvessel1
AT SEA
11/28/2021 23:09
Ping: 1 hrs

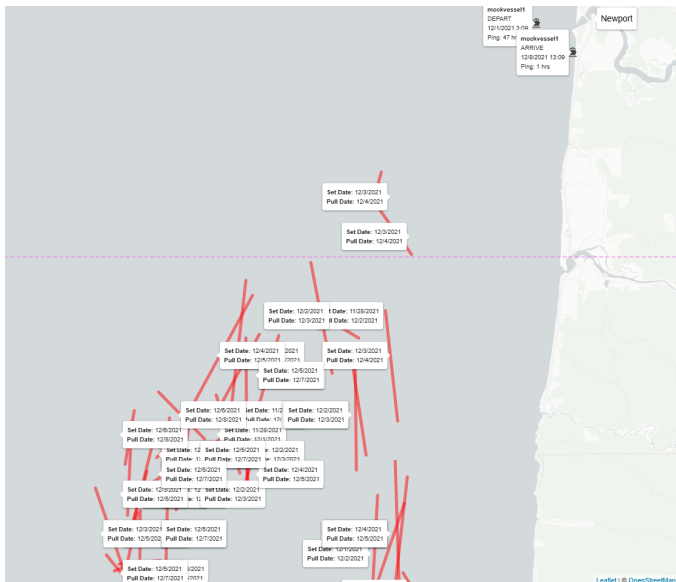
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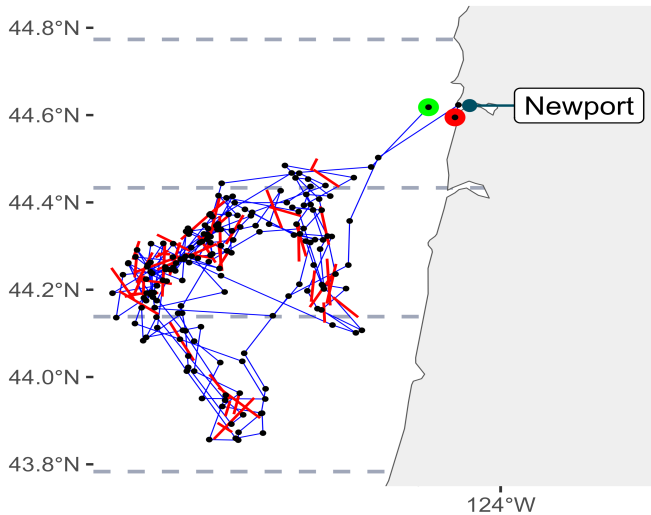
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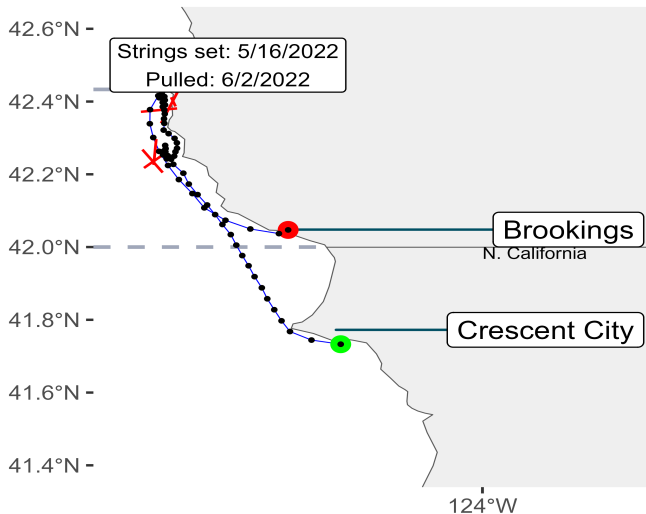
DATA EXAMPLE



DATA EXAMPLE



DATA EXAMPLE



DATA SUMMARY

Summary Statistics by Evisceration Status

	Untreated Areas		Treated Areas		Diff. in Means	p
	Mean	Std. Dev.	Mean	Std. Dev.		
Price per Pound (\$/lb)	4.77	1.63	4.45	1.38	-0.31	<0.01
Revenue per Vessel (\$/week)	4785.44	8396.99	6854.74	11409.66	2069.30	<0.01
Landings per Vessel (lbs/week)	1333.37	2893.36	2045.04	3792.38	711.66	<0.01
Total Landings (lbs)	27719.14	91316.32	28569.02	76905.82	849.87	0.86
Average Pots per Vessel	161.48	116.56	165.17	117.28	3.69	0.60
Catch per Unit Effort (lbs/pot)	6.07	6.41	9.31	10.35	3.24	<0.01
Number of Active Vessels	10.48	10.98	8.99	7.72	-1.49	<0.01
Share of Large Vessels (%)	0.32	0.29	0.23	0.25	-0.09	<0.01

EMPIRICAL STRATEGY

Baseline DiD specification:

$$Y_{isw} = \gamma \text{Evis}_{isw} + \beta X_{isw} + \alpha_i + \delta_{sw} + \varepsilon_{isw}$$

- α_i : area fixed effects absorb time-invariant productivity and infrastructure differences
- δ_{sw} : season-week fixed effects absorb market-wide supply, demand, and weather conditions
- Treatment effects identified from within-area temporal variation

EMPIRICAL STRATEGY: EVENT STUDY

Event-study model takes the form of:

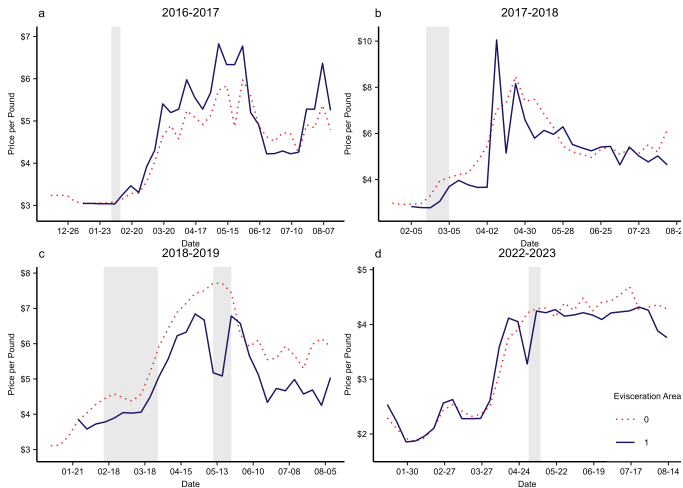
$$Y_{ist} = \left(\sum_j \gamma_j D_{i,t-j} \right) + \beta X_{ist} + \alpha_i + \delta_{st} + \epsilon_{ist}$$

Weights (Wing et al., 2024)

$$Q_{sa} = \begin{cases} 1, & \text{if } D_{sa} = 1, \\ \frac{N_a^D / N_{\Omega_k}^D}{N_a^C / N_{\Omega_k}^C}, & \text{if } D_{sa} = 0 \end{cases}$$

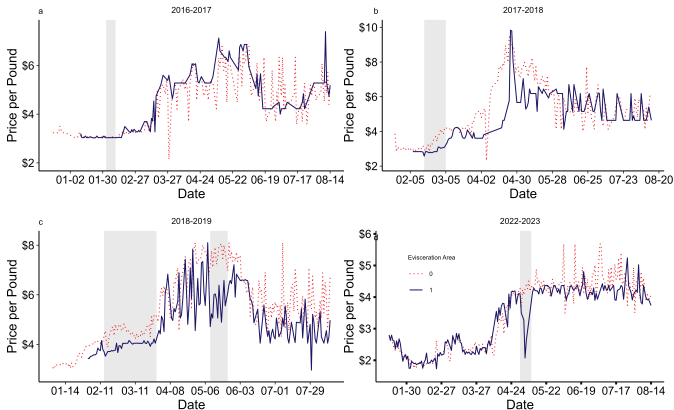
- Groups weighted by each sub-experiment's share of treatment-induced variance

PARALLEL TRENDS: SEASON-WEEK



Weekly aggregation shows apparent parallel trends in most seasons

PARALLEL TRENDS: DAILY



At daily frequency, the apparent pre-trend in panels (c) and (d) resolves, treatment and control track closely until the order is announced, then diverge sharply. Weekly figure conflates pre and post-order observations within the same bin when orders fall mid-week.

PARALLEL TRENDS

In all seasons, prices between treatment and control groups exhibit similar trends before policy.

Post-treatment divergence is sharp and visible in 2018–19 and 2022–2023. Recovery follows order expiration.

Variation in magnitude and timing across seasons is consistent with differences in market conditions and order duration.

PRICE EFFECTS

Dependent Variable: Price per Pound		
	(3)	(5)
Evisceration Order	-0.516*** (0.150)	-0.748*** (0.184)
Num.Obs.	6143	6143
R2	0.906	0.907
R2 Adj.	0.893	0.894
Season-Week FE	Yes	Yes
Area FE	Yes	Yes
Composition Controls	No	Yes
Derby Controls	No	Yes

Preferred specification: Column (5)

$$\hat{\gamma} = -\$0.748 \text{ per lb} \quad (p < 0.001)$$

Approximately a 10–15% reduction relative to the sample mean of \$4.77/lb.

Interpretation:

- Live/whole crab market eliminated
- Product shifted into cooked/processed channels
- Effect is stable across specifications

PRICE EFFECTS: EVENT STUDY

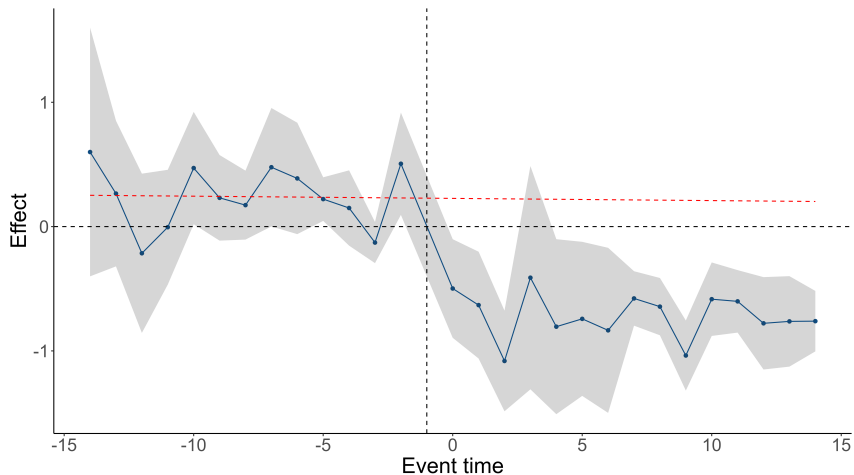


Figure presents event study estimates following the stacked format as presented in Cengiz et al. (2019); Wing et al. (2024). The event time is a 14 day window around the policy. Red dashed line is pre-trend line.

PRICE EFFECTS: EVENT STUDY

Prices decline by \$1.01–\$0.69 per pound within 2–3 days of order announcement.

- Pre-period coefficients are show no clear trends before treatment — no anticipatory price
- Price effect is immediate, consistent with markets rapidly adjusting to the loss of premium live channels

REVENUE EFFECTS

	(1)	(2)	(3)	(4)	(5)
Evisceration Order	2655**	3064***	2541**	2281*	-3746***
R2	0.00	0.03	0.72	0.73	0.75
FE/Controls	No	Partial	Yes	Yes	Yes

Columns 1–4 (without derby controls) show a positive revenue effect. Column 5 reverses this to –\$3,746 ($p < 0.001$). The earlier specifications confound evisceration timing with naturally high-revenue opening weeks.

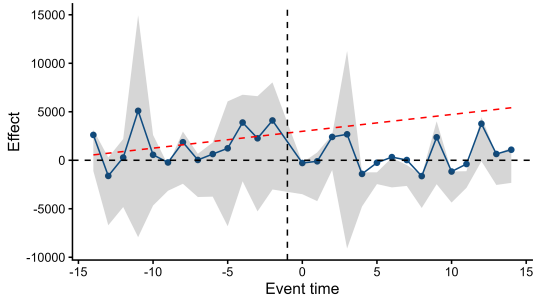
THE PUZZLE

Despite price reduction, vessels do not exit, reduce effort, or relocate to unaffected areas.

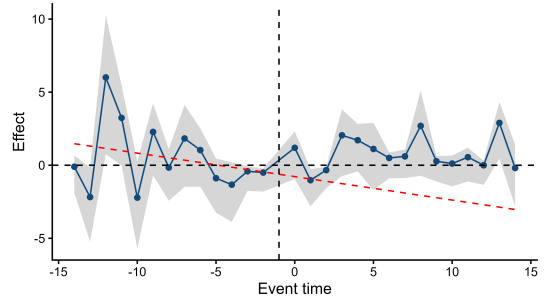
The derby structure potentially explains why. Vessels have pre-positioned gear, established buyer relationships, and local ecological knowledge, and with gear that cannot be quickly redeployed in response to a constraint averaging 12–18 days.

Vessels do not drastically change behavior in response to price effect.

EFFORT AND PARTICIPATION: NO ADJUSTMENT



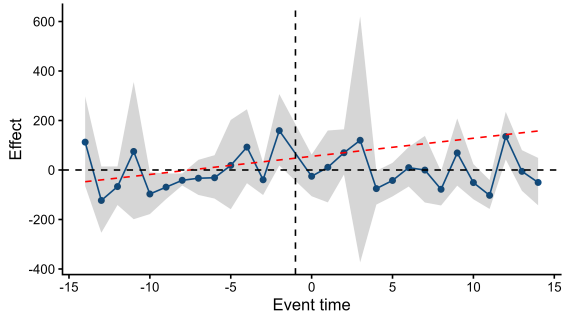
(c) Figure present event study estimates on vessel landings



(d) Figure present event study estimates on vessel counts

Neither landings per vessel nor the number of active vessels changes significantly during evisceration periods. Both series show no systematic pattern, and the estimates are statistically insignificant.

NO ADJUSTMENT: POT USAGE



Pot usage, the primary measure of fishing effort, shows no systematic response to evisceration orders. Point estimates fluctuate around zero throughout the event window.

Catch per unit effort is similarly flat, confirming that fishing productivity is unaffected.

Vessels are constrained by pre-positioned gear that cannot be easily relocated in response to a short-lived policy shock.

SPILLOVERS: LIMITED SPATIAL RELOCATION

Logbook-data : Border areas adjacent to eviscerated zones experience a statistically insignificant 1.4% increase in vessel counts during evisceration; pure control areas see a 9.9% increase, also insignificant.

Limited relocation? Three possible mechanisms: orders are short-lived (12–18 days), making relocation to unfamiliar grounds unprofitable (Girardin et al., 2017); fishing grounds are not perfect substitutes; and if effort is pre-positioned at season opening, the cost of moving gear is high relative to the duration of the shock.

SPILLOVERS: LOGBOOK

Effect of Evisceration Orders on Vessel Movement

	(1)	(2)	(3)	(4)
Evis	-0.007 (0.018)	-0.068 (0.044)	-0.820** (0.301)	-0.065 (0.120)
Neighbor Evis			0.920** (0.367)	1.095** (0.415)
Evis × Neighbor Evis				-1.012** (0.421)
Num. Obs.	6,143	6,143	6,143	6,143
R ²	0.000	0.116	0.316	0.349
R ² Adj.	0.000	0.025	0.245	0.282
Season–Week FE	No	Yes	Yes	Yes
Area FE	No	Yes	Yes	Yes

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

HETEROGENEITY: FLEET AND MARKET STRUCTURE

The average treatment effect raises two questions about what is driving the price result.

Is the policy effect uniform across vessel types?

The fleet operates at very different scales. The fleet is comprised of vessels who differ greatly in terms of size and capacity. Some vessels are under 30 ft while others are over 50 feet and can hold over 100,000 pounds of crab.

Can port structure explain the price effect?

Ports vary substantially in processing capacity. Large processors can route eviscerated product into alternative channels; smaller buyers cannot. If buyer structure is driving prices, the effect should be larger in ports dominated by smaller buyers.

HETEROGENEITY: VESSEL TYPE RESULTS

Landings	
Evis	-792.936** (284.755)
Medium	-922.561*** (208.396)
Small	-1806.088*** (266.266)
Evis · Medium	74.251 (424.047)
Evis · Small	23.911 (711.833)
R ²	0.494

Revenue	
Evis	-2948.743** (1012.801)
Medium	-2959.589*** (666.092)
Small	-5835.692*** (828.743)
Evis · Medium	91.038 (1596.958)
Evis · Small	-517.503 (2576.784)
R ²	0.471

- Evisceration reduces landings by 793 lbs/week and revenue by \$2948.74 for large vessels (baseline category)
- Interaction terms for medium and small vessels are statistically insignificant

HETEROGENEITY: BUYER STRUCTURE AND PRICES

	(4)
Evis	-1.051*** (0.151)
Port-Large Dealer	-0.304*** (0.074)
Evis $\times$ Port-Large Dealer	0.654*** (0.142)

Evisceration reduces prices by approximately 50 cents per pound on average across ports.

The effect is substantially larger in ports dominated by smaller buyers. Access to large processors seems to attenuate the effect

DOMOIC ACID TESTING PROVIDE INFORMATION?

Bivalves are early indicators of domoic acid events in crab, since they are a primary food source for Dungeness crab. If vessels respond to bivalve contamination signals before an order is announced, we would expect anticipatory behavioral adjustment.

Domoic acid violations in bivalves do not produce significant anticipatory responses. The DA violation in crab indicator is similarly insignificant.

Vessels respond to the regulatory announcement, not to the underlying environmental signal. The policy itself, not the ecological event, is the relevant information event for fishing behavior.

DOMOIC ACID TEST RESULTS

Dependent Variable= Moved String(s) to Untreated Area						
	(1)	(2)	(3)	(4)	(5)	(6)
Evis	0.103*** (0.001)	0.122*** (0.039)	0.120*** (0.039)	0.120*** (0.039)	0.119*** (0.039)	0.040 (0.065)
Medium		0.000 (0.002)	0.000 (0.002)	0.000 (0.002)	-0.000 (0.002)	-0.000 (0.002)
Small		-0.001 (0.002)	-0.001 (0.002)	-0.001 (0.002)	-0.001 (0.002)	-0.002 (0.002)
Medium · Evis		-0.023 (0.045)	-0.025 (0.045)	-0.025 (0.045)	-0.025 (0.045)	0.019 (0.047)
Small · Evis		-0.130*** (0.039)	-0.131*** (0.038)	-0.130*** (0.038)	-0.130*** (0.038)	-0.055 (0.056)
DA Violation Crab			0.036 (0.035)	0.036 (0.035)	0.035 (0.035)	0.040 (0.035)
DA Violation Bivalve				-0.010 (0.006)	-0.010 (0.006)	-0.009 (0.006)
Num.Obs.	80337	80337	80337	80337	80336	80336
R2 Adj.	0.059	0.079	0.080	0.080	0.080	0.089
Season-Day FE	No	Yes	Yes	Yes	Yes	Yes
Vessel FE	No	Yes	Yes	Yes	Yes	Yes
Controls	No	No	No	No	Yes	Yes

* p < 0.1, ** p < 0.05, *** p < 0.01

Robust standard errors in parentheses clustered by vessel

PUTTING THE RESULTS TOGETHER

What the policy does

- Reduces ex-vessel prices 10%
- Reduces revenue
- Preserves landings and effort

What the policy does not do

- Trigger vessel exit
- Cause significant spatial reallocation of effort
- Create anticipatory behavioral responses

The policy operates through prices

Vessels absorb the regulatory shock entirely through lower revenues. Relative to the alternative of a full closure, the policy preserves economic activity across the fleet

LIMITATIONS AND FUTURE DIRECTIONS

Compositional bias. The aggregate DiD does not fully address differential vessel participation during evisceration. Sant'Anna and Xu (2026) show this can bias estimators even under standard identifying assumptions. Addressed partially through the vessel-type disaggregation; full treatment remains future work.

Reduced-form scope. The analysis identifies broad patterns and average treatment effects. Structural models, such as discrete choice frameworks, would support counterfactual simulation and welfare analysis (Smith and Wilen, 2003; Girardin et al., 2017).

Logbook matching. Matched logbook-ticket data underreports total landings. Sample means of ex-vessel prices between matched and full ticket datasets are not statistically different, suggesting the matched sample is representative, but the volume mismatch introduces measurement uncertainty.

CONCLUSION

Evisceration orders reduce ex-vessel prices by roughly ten percent without triggering a meaningful adjustment in fishing effort, vessel participation, or spatial behavior, consistent with high switching costs in a derby fishery where the compressed season and pre-positioned gear limits firms' ability to respond to short-lived regulatory shocks.

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